

CompTIA Network+ Project

As a young, trained network engineer, you have been tasked with designing and implementing the New Horizons Network. Studied the requirements carefully, then designed and implemented the network based on the company's needs.

New Horizons Nigeria is an IT training center that has three buildings (Building A, B and C) situated at Nos. 5 and 6 Babatola Drive off Awolowo Way, Ikeja. The buildings are 50 meters and 10 meters apart.

Building A has three floors with two departments in each for example;

First floor- (Reception and Counselling Department-7 users expected,

Second floor- (Manager office and Technical Department-10 users expected,

Third floor- (Conference room and the executive office -9 devices expected)

The requirement during the design and implementation are has follow;

1. There should be three routers connecting each floor (all placed in the server room in technical department).
2. All routers should be connected to each other using serial DTE cable.
3. The network between the routers should be 192.168.1.0/30, 192.168.2.0/30 and 192.168.3.0/30.
4. Each floor is expected to have one switch (placed in the respective floor).
5. Each department is expected to be in different VLAN with the following detail and use this network address, **192.168.100.0/28**.

1st Floor;

Reception VLAN 10, Use the second subnets of the network address given

Counselling VLAN 20, Use the third subnets of the network address given

2nd Floor;

Manager office VLAN 30, Use the sixth subnets of the network address given

Technical VLAN 40, Use the fifth subnets of the network address given

3rd Floor;

Conference room VLAN 50, Use the tenth subnets of the network address given

Executive office VLAN 70, Use the last subnets of the network address given

6. Use RIP version 2 as the routing protocol to advertise routes.
7. All devices in the network are expected to obtain IP address dynamically with their respective router configured as the DHCP server.
8. All the devices in the network are expected to communicate with each other.
9. Configure port security to the second-floor switch to allow only Technical PCs to access port fa0/1-10 (use manual method to obtain mac-address with violation mode of shutdown.)
10. Please note that each floor is expected to have a wireless network connection for laptops and phones for visitors.

Building B

This consists of training classes, cooperating offices, and labs. All the devices in the labs, training classes, and cooperating offices are expected to share resources and communicate with each other. Note, you are expected to configure the network devices to provide end-to-end connectivity. Hint: use the OSPF routing protocol here. You need to provide addresses from a DHCP server to hosts that aren't on the same LAN as the DHCP server. You can configure your router interface to relay or forward the DHCP client requests. Use the second, third, and sixth subnets of 192.168.10.0/27, network addresses, while the first subnet should be used for DHCP server relay, and make sure you have at least five hosts in each, except for the cooperating offices, which should have a minimum of four end users.

Building C

This consists of administrative, HR, and account staff. The PCs of the staff are distributed in offices, and it is expected they will share some networking equipment. Hint: use of router on a stick inter-VLAN routing is expected and make sure they obtained addresses manual, and note that administrative and HR are on the same subnet and VLAN, while the account is on a different VLAN and subnet. Use the second and third subnets of this network address, 192.168.100.0/26, and also, the third, sixth, and last valid hosts for the second subnet, while for third subnet, use the first, second, fourth, and last valid hosts.

Tasks:

1. Your task is to plan, design, and prototype the network topology for New Horizons Nigeria, an IT training center, using Cisco Packet Tracer.
2. Configure the network in the packet tracer with appropriate settings (Ensure the administrative configuration is done on your Routers and Switches) to achieve the connectivity and functionalities specified in the requirements.
3. Make sure you use appropriate cable connection to connect end user devices.